Alejandro Quílez Asensio

Data Platform Architect · Azure · Geospatial Platforms

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PROFILE

Data architect with an engineering background spanning Azure infrastructure, data platforms, and software design. Currently designing, building, and managing the architecture of a geospatial platform for a public agency handling data at petabyte scale. Earlier work includes banking decision systems, regulated insurance data pipelines, and telecom ingestion platforms.

CURRENT EXPERIENCE

Data Architect — Space sector company

May 2026 – Present · Public-sector geospatial platform

- Design, build, and manage the architecture of an Azure platform for a public agency handling data at petabyte scale.
- Design networking and Microsoft Entra ID components within infrastructure built around ENS High requirements.
- Implement medallion data architecture and distributed HPC workloads on Azure Batch.
- Design and implement MLOps components and business databases as part of the platform.
- Develop an internal library that acts as a shared control plane, connecting the work of multiple teams.
- Write architecture decision records (ADRs) and architecture documentation to support implementation and ongoing platform evolution.

Azure · Azure Batch · HPC · Microsoft Entra ID · MLOps · Medallion architecture

PREVIOUS EXPERIENCE

Senior Data Engineer / Software Engineer — Knowmad Mood

January 2025 – May 2026 · Banking

- Refactored a core decisioning system into modular, testable Python services using async and event-driven architecture.
- Increased daily business capacity from 400 to 500 requests (+25%), improving execution and end-to-end traceability.
- Worked on system architecture, testing strategy, observability, and CI/CD for production delivery.

Python · Async · Event-driven architecture · Azure · CI/CD

PREVIOUS EXPERIENCE, CONTINUED

Senior Data Engineer — Knowmad Mood

April 2023 – December 2024 · Insurance

- Built and improved Azure Databricks and PySpark pipelines in a regulated insurance environment, reducing execution from hours to minutes.
- Designed fact and dimension models for analytical reporting and introduced data validation and quality checks.
- Implemented CI/CD for data workflows and collaborated with business and analytics teams.

Machine Learning Engineer / Data Engineer — Capgemini

February 2021 – April 2023 · Telecom

- Designed ingestion architecture processing approximately 2 TB per day across more than 80 tables.
- Automated data lake maintenance and worked on machine learning deployments and CI/CD processes.
- Acted as a technical reference and mentored junior engineers on data and machine learning platforms.

Innovation Engineer — Capgemini

October 2020 – February 2021

- Designed backend architectures for service prototyping on Google Cloud Platform.
- Developed APIs and integrations to automate data access and internal workflows.

TECHNICAL EXPERTISE

Architecture: platform design, ADRs, architecture documentation, shared control planes, software design patterns.

Azure foundations: networking, Microsoft Entra ID, infrastructure as code with Terraform, security requirements.

Data and compute: geospatial platforms, Azure Batch, HPC, Databricks, Spark / PySpark, Delta Lake, medallion architecture, SQL.

Engineering and delivery: Python, async services, event-driven architecture, MLOps, MLflow, CI/CD, Azure DevOps, Docker.

EDUCATION

BSc in Telematics Engineering — University of Valencia

January 2016 – January 2021 · Valencia, Spain